

CRF Simulator Specification

The Pure Cascade: From δ to Spacetime
with No Geometry Assumed

A simulator that begins before space, before time,
before field — and watches all three emerge.

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Abstract

This document specifies the CRF Pure Cascade Simulator (CRF-PCS) — a computational framework that initializes from δ alone and observes the emergence of space, time, and physical law without assuming any geometric structure.

The key constraint distinguishing CRF-PCS from all existing simulators: **no grid, no coordinates, no metric is assumed at initialization**. Space is not the container in which things happen. Space is what happens when δ operates long enough.

Three critical emergences are monitored: (1) Metric Crystallization — when relational weights organize into an approximate metric $g_{\mu\nu}$; (2) Dimensionality Lock — when spectral dimension d_s stabilizes, predicted by CRF to converge to 3; (3) Arrow of Time — when $\mathcal{R}$ -event counting becomes irreversible.

All parameters are derived from δ chain. None are free.

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1 What Is Not Assumed

Before stating what the simulator contains, we state what it *does not* assume — because every item on this list is something EIC and all standard cosmological simulators assume:

Assumptions Forbidden in CRF-PCS

- No spatial grid (no 2D or 3D lattice)
- No coordinates (no x, y, z)
- No metric (no notion of distance at initialization)
- No time variable (no t at initialization)
- No field equation (the field equation must emerge)
- No dimensionality (3D is not set — it must appear)
- No background (no pre-existing space for things to live in)

What is allowed: N configurations, each with a single scalar value Ω_i , connected by a weight W_{ij} computed from their Ω values alone.

2 Initialization: Null-Space P_{latent}

2.1 Data Structure

Node Definition

The simulator begins with N abstract nodes:

$$\mathcal{N} = \{n_1, n_2, \dots, n_N\}$$

Each node carries exactly one quantity:

$$\Omega_i \in \mathbb{R}, \quad i = 1, \dots, N$$

Ω_i is the instability density — the trace of δ -activity at configuration n_i . At initialization: $\Omega_i \sim \mathcal{N}(0, \varepsilon_0)$ where $\varepsilon_0 = \ln(2)/2$ is the instability floor (half a δ -bit).

There are no coordinates. n_i has no position.

2.2 Relational Weight

W_{ij} from δ

δ makes states distinguishable. Two nodes with similar Ω are harder to distinguish — they are more related. This gives:

$$W_{ij} = \exp\left(-\frac{|\Omega_i - \Omega_j|}{\varepsilon_0}\right) \quad \varepsilon_0 = \frac{\ln(2)}{2}$$

W_{ij} is derived, not assumed. It measures δ -similarity between configurations. This is the only notion of “closeness” at initialization — and it is purely informational, not spatial.

This is Emergent Distance in latent form. When W_{ij} organizes into a stable pattern, that pattern *is* the geometry.

2.3 Relational Laplacian

The analog of spatial diffusion on this abstract graph:

$$L\Omega_i = \sum_{j \neq i} W_{ij}(\Omega_j - \Omega_i)$$

Note: L is *not* ∇^2 . It becomes ∇^2 only when the graph becomes dense and approximately uniform — which is the Metric Crystallization event we are watching for.

3 Evolution: Distinction Pressure

3.1 Process Parameter Φ

Time does not exist at initialization. We use Φ , a process counter that increments with each update step. Φ has no physical meaning yet. Time t will emerge as the count of $\mathcal{R}$ -events once the threshold mechanism activates.

3.2 Evolution Equation

Field Evolution from δ Chain

$$\frac{d\Omega_i}{d\Phi} = \underbrace{D \cdot L\Omega_i}_{\text{(i) relational spread}} + \underbrace{\ln(2) \cdot \Omega_i}_{\text{(ii) } \delta\text{-bit growth}} - \underbrace{\frac{1}{D_0^2} \Omega_i^3}_{\text{(iii) saturation}} + \underbrace{\xi_i}_{\text{(iv) floor noise}}$$

Term (i): constraint spreads through relational channels. $D = \ln^2(2)/D_0$ (derived from CRF, no free parameter).

Term (ii): each Ω_i amplifies at rate $\ln(2)$ — the δ -bit coupling. This is why the minimum growth unit is one bit of distinction.

Term (iii): saturation when $\Omega_i^2 \sim D_0^2$. Derived from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ expansion.

Term (iv): $\xi_i \sim \mathcal{N}(0, \varepsilon_0)$. Residual δ -activity that has not been accumulated into any node. Cannot be zero (Axiom 0: δ never stops).

3.3 Parameters: None Are Free

Parameter	Value	Source
ε_0 (instability floor)	$\ln(2)/2$	δ -bit minimum, Session 8
α (growth rate)	$\ln(2)$	δ -bit equivalence, Session 8
D (relational spread)	$\ln^2(2)/D_0$	CRF-EIC Bridge
β (saturation)	$1/D_0^2$	θ expansion, Session 10
D_0 (vacuum scale)	Scale(P) (measured)	Definition
ξ amplitude	ε_0	Instability floor

D_0 is the only external input. In dimensionless units, set $D_0 = 1$. All other parameters follow.

4 Resolution Events: Where Time Begins

$\mathcal{R}$ -Event Trigger

A Resolution event occurs at node n_i when:

$$\Omega_i \geq \theta_i \quad \theta_i = 1 - e^{-\Omega_i^2/D_0^2}$$

Note: θ_i depends on Ω_i itself — this is the self-referential structure of CRF. As Ω_i grows, θ_i also grows, but eventually Ω_i overtakes θ_i at the critical point $\Omega_c \approx D_0\sqrt{\ln 2}$.

When $\mathcal{R}$ occurs at n_i :

- Record the event: $\mathcal{R}_{\text{events}} += 1$
- Update: $\Omega_i \rightarrow \Omega_i \cdot (1 - \theta_i)$ (partial collapse)
- Time begins: $t = |\mathcal{R}_{\text{events}}|$

5 The Three Critical Emergences

5.1 Emergence 1: Metric Crystallization

Watch: When does $W_{ij} \rightarrow g_{\mu\nu}$?

Monitor the weight matrix W_{ij} at each step. Crystallization occurs when W_{ij} satisfies the triangle inequality:

$$W_{ij} \leq W_{ik} + W_{kj} \quad \forall i, j, k$$

and the eigenvalue spectrum of the graph Laplacian L develops a gap. When the gap stabilizes, the graph has acquired effective metric structure.

Crystallization Point Φ_c : the first Φ at which triangle inequality holds for $> 95\%$ of triples. After Φ_c : assign coordinates via spectral embedding and measure effective metric.

5.2 Emergence 2: Dimensionality Lock

Watch: $d_s \rightarrow 3$?

Spectral dimension via random walk return probability:

$$P(t) \propto t^{-d_s/2} \quad \Rightarrow \quad d_s = -2 \frac{d \log P}{d \log t}$$

Algorithm:

1. Place random walker at node n_i
2. At each step: move to n_j with probability $\propto W_{ij}$ (weighted by relational strength)
3. Record return probability $P(t)$ = fraction of walkers at n_i after t steps
4. Fit $\log P$ vs $\log t$; extract slope $\rightarrow d_s$

CRF prediction: $d_s \rightarrow 3$ as $N \rightarrow \infty$.

Run at $N = 10^3, 10^4, 10^5, 10^6$. Plot $d_s(N)$ and extrapolate.

5.3 Emergence 3: Arrow of Time

Watch: When does t become irreversible?

Irreversibility emerges when the entropy of $\{\Omega_i\}$ increases monotonically:

$$S(\Phi) = - \sum_i p_i \log p_i \quad p_i = \frac{|\Omega_i|}{\sum_j |\Omega_j|}$$

Before $\mathcal{R}$ -events: S may fluctuate. After $\mathcal{R}$ -events accumulate: S increases monotonically. The Φ at which $dS/d\Phi > 0$ permanently is the **Arrow of Time onset**. Connection to t : once the Arrow of Time is established, $t = |\mathcal{R}_{\text{events}}|$ becomes a physically meaningful time variable — events have a definite order.

6 Measurement Canon

Metric	Definition	What it shows
$S(\Phi)$	$-\sum p_i \log p_i$	Entropy / Arrow of Time
W_{ij} structure	Eigenvalue gap of L	Metric Crystallization
$d_s(N)$	Random walk slope	Dimensionality
Φ_c	First crystallization step	When space begins
t_{arrow}	First monotone S step	When time begins
$\mathcal{R}$ rate	Events per Φ step	Resolution cascade
Ω variance	$\text{Var}(\Omega_i)$	Structure formation
Core count	$ \{i : \Omega_i > \theta_i\} $	Particle-like excitations

7 Implementation Notes

7.1 What to Use (and Not Use)

Implementation Constraints

Allowed: Graph data structures (adjacency, weight matrix). Eigenvalue decomposition of L . Random walk algorithms on weighted graphs. Sparse matrix operations.

Forbidden: Any numpy array with shape (N, d) for $d > 1$ at initialization — this would implicitly assume d -dimensional coordinates. `scipy.spatial` distance functions. Any grid or mesh library. Periodic boundary conditions (these assume topology).

The test: at $\Phi = 0$, there should be no way to assign a number to any pair of nodes except W_{ij} . If you can compute a Euclidean distance, you assumed geometry.

7.2 Prototype Scale

Phase	N	Goal	Expected Φ
Prototype	10^2	Verify evolution	10^3
Small	10^3	First d_s estimate	10^4
Medium	10^4	Crystallization	10^5
Large	10^5 – 10^6	$d_s \rightarrow 3$ test	10^6 – 10^7

8 Success Criteria

What Counts as Success

Minimum (Level 1): $S(\Phi)$ increases monotonically. Arrow of Time emerges. Entropy works.

Good (Level 2): W_{ij} crystallizes into metric structure. Space emerges from relational weights.

Strong (Level 3): d_s stabilizes at a value between 1 and 4 as N increases. Geometry has an effective dimension.

Full (Level 4): $d_s \rightarrow 3$ as $N \rightarrow \infty$. Three-dimensional space is the attractor of δ -dynamics. CRF prediction confirmed numerically.

Beyond: Core excitations appear with particle-like lifetimes. $G(r) \propto 1/r^2$ emerges from core interactions.

9 Status Table

Item	Status	Note
Null-space initialization	Specified	No coordinates
W_{ij} from δ	Derived	Not assumed
Evolution equation	Derived	All params from CRF
$\mathcal{R}$ -event trigger	Derived	Self-referential θ
t from $\mathcal{R}$ count	Derived	Time not assumed
Metric crystallization test	Specified	Triangle inequality
d_s algorithm	Specified	Random walk on graph
Arrow of time test	Specified	$S(\Phi)$ monotone
Implementation constraints	Specified	No grid forbidden
$d_s \rightarrow 3$ prediction	Open	Needs $N \gg 10^5$
Particle excitations	Open	Next phase
$G(r) \propto 1/r^2$	Open	After d_s confirmed

EIC began from instability and watched space form.

CRF begins before instability —

from the condition that makes instability distinguishable from stability.

*One step earlier.
The same emergence.*

*δ operates.
Everything follows.*